Study Questions: Cognition — Shared Understanding Without Words

1. Explain what **distributed knowledge** means in the context of a crochet circle. How does the group's collective knowledge exceed the sum of individual members' knowledge?

2. Describe a specific problem-solving scenario in a crochet circle. Trace the process through the five stages: problem presentation, hypothesis generation, evidence evaluation, solution convergence, and validation.

3. How does the question "Has anyone worked with this yarn before?" function as a **cognitive tool**? What makes it efficient in Active Inference terms?

4. What is a **shared mental model**? Give an example of a shared mental model that your circle (real or imagined) has developed about difficulty levels.

5. How does a circle's shared understanding of what counts as "easy" versus "advanced" differ from a single crocheter's private assessment? What makes the shared model more reliable?

6. Describe how **nonverbal cognition** works in a crochet circle. Give an example of a complex understanding being communicated through gesture and demonstration without words.

7. Why is nonverbal cognition particularly powerful in crochet compared to domains that rely more on abstract reasoning? What about crochet makes embodied demonstration so effective?

8. When multiple circle members offer different hypotheses for why a fabric is curling, how does the group perform **collective Bayesian inference**? What serves as the "evidence" that distinguishes between hypotheses?

9. How do **aesthetic norms** function as shared mental models in a circle? What happens when a new member's aesthetic differs from the group's established norms?

10. Describe the concept of **process knowledge** in a circle — shared heuristics like "when in doubt, swatch first." How do these heuristics develop over time?

11. When a member presents a finished project to the circle, the group's response constitutes **collective evaluation**. How does this group evaluation differ from self-evaluation? What additional information does it provide?

12. A new member joins the circle and does not yet share the group's mental models. How does the process of absorbing these shared models work? Is it explicit or implicit?

13. How does **material knowledge** — the collective understanding of how different yarns and tools behave — accumulate in a circle? What role do individual experiences (both successes and failures) play?

14. Consider two circles: one that has been solving problems together for years and one that just formed. How do their collective cognitive capacities differ?

15. In Active Inference, cognition involves updating beliefs based on prediction error. Give a crochet circle example where the group's prior belief about a technique was updated by new evidence.

16. How does **physical co-presence** — being in the same room — enhance the circle's collective cognition compared to communicating online or through written patterns?

17. Explain how a circle member's partial knowledge becomes more valuable in the group context than in isolation. What mechanisms allow partial knowledge to combine with others' partial knowledge?

18. When the circle develops a shared model about how a new yarn behaves, is this model stored in any one person's head, or is it distributed? How is it maintained and accessed?

19. Sometimes the circle reaches a wrong consensus — a shared belief that turns out to be incorrect. How does this collective prediction error get corrected?

20. Reflect on the phrase "the circle makes every member smarter than they would be alone." What does this mean in Active Inference terms? What cognitive capacities does group membership provide that solitary practice does not?